3.7.2 Collisional Verlet Algorithm for Hard Spheres

A method was given in [184] based on a modified collision procedure which greatly reduces the error growth.

The idea is to make use of the quadratic

$$\boldsymbol{Q}(t) = \boldsymbol{q} + t\boldsymbol{M}^{-1}\boldsymbol{p} - \frac{t^2}{2}\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}) \tag{3.18}$$

which represents the position vector obtained from a Verlet step of size t. Note that this defines quadratic paths for all particles in the system. It is then possible to calculate the collision times by solving equations

$$\|\boldsymbol{Q}_i(t) - \boldsymbol{Q}_j(t)\| = \sigma_i + \sigma_j, \qquad i \neq j. \tag{3.19}$$

A step can then be taken to the first point of subsequent collision, with positions updated using the quadratic and momenta adjusted according to the Verlet map combined with $\mathscr{R}_c$. In this way, all steps taken are Verlet steps so the order of accuracy is two. Effectively, this is a Verlet method with variable timestep chosen to match collision times. A maximum outer stepsize $h_{\max}$ is incorporated to prevent too-large steps from being taken (in case collisions are infrequent). We write the method here in a slightly different form than in [184]: both the stepsize taken and the coordinates are viewed as variables.

Collisional Verlet Algorithm (CVA) [Single Step]

[computes h (the timestep) and $(\boldsymbol{Q},\boldsymbol{P})$ *given a starting point* $(\boldsymbol{q},\boldsymbol{p})$*]*

Calculate τ_c the time of next collision from the Verlet paths (quadratics) of (3.18) using collision conditions (3.19).
If $\tau_c < h_{\max}$, then

$h := \tau_c$
$(\boldsymbol{Q},\boldsymbol{P}) := \mathscr{R}_c\mathscr{G}_h^{\text{Verlet}}(\boldsymbol{q},\boldsymbol{p})$

else

$h := h_{\max}$
$(\boldsymbol{Q},\boldsymbol{P}) := \mathscr{G}_h^{\text{Verlet}}(\boldsymbol{q},\boldsymbol{p})$

An alternative to Verlet could be used for propagation between collisions, but it is likely that the collision computations would then become more costly.

If the Verlet method is used, then it turns out that the insertion of (3.18) into (3.19) results in a quartic polynomial that must be solved for each particle pair. Care must be taken in solving these quartics to ensure that roots are not missed. Using hash tables, neighbor lists and similar methods (like those used to compute non-bonded forces), the number of computations can be greatly reduced. Importantly, the calculations needed are all explicit and finite (no iteration is needed).